Frame transport for framed formal monodromy: the lemma, its hypotheses, and the receiver problem

Fifth version, closing the coalesced case

Memo for a reader of *Irrationality of Cubic Threefolds after One Stabilization*

15 August 2026

Abstract

Proposition 4.7 of the draft transports framed formal monodromy across the comparison isomorphisms of Iritani and Iritani–Koto, and the transport step itself is asserted in one sentence. This memo states and proves the missing lemma over a differential field, with its hypotheses made explicit; shows that those hypotheses are jointly unsatisfiable over the coefficient receiver the draft builds, so that the lemma as proved has no verified instance in the application; corrects the receiver; and reduces the remaining question to one inequality that depends on a normalization the draft is free to choose. The inequality fails under the draft's current instruction to take the separating weight parameter large, and holds at the cubic endpoint under the smallest admissible choice. Section 4 examines an iterated receiver that would satisfy the inequality trivially, checks it against the sources, and finds that it needs a semipositivity hypothesis on the ambient variety which weak factorization does not supply.

The second version of this memo proposed the framing theory of Hinault, Yu, Zhang and Zhang as the probable repair, on the ground that its decomposition gauge is regular in the loop coordinate. Three compatibility checks against the pinned sources have since been run, and the route does not survive them; Section 5 now reports what those checks found. Two consequences are worth more than the route was. First, no receiver is needed for the comparison isomorphisms at all: Iritani's Ψ and Iritani–Koto's Φ are polynomial in the loop coordinate with inverses in the same ring, so the transport theorem below applies to them with $\Gamma = 0$. Second, the diagnosis of the second version was wrong — the pro-Laurent bulk gauge is not the draft's own choice but Iritani–Koto's own fundamental solution, which the sources meet and handle by Birkhoff factorization. What is left is a single statement about one bulk displacement, recorded in Section 5. Section 6 then attacks that statement directly through the flatness identities. It proves that multiplicity-one blocks carry no framed monodromy, which is unconditional and leaves only coalesced blocks in play, and writes down the exact evolution of the block's leading operator. It does *not* reduce the coalesced case: the third version of this memo claimed such a reduction and was wrong twice over, once in dropping a term from the off-block Sylvester equation and once in modelling a nonsemisimple block by its compressed pair. Both errors, and the cubic 2×2 example that is the regression test for any repair, are recorded there.

Section 7, new in this version, closes the coalesced case for a block of the cubic's type, and with it the residual gap for the cubic ambient summand. The device is to shear the block before deforming it. Three statements are proved over the formal even bulk germ, from the flatness identities and nothing else: the double eigenvalue never splits, because $\det N$ satisfies $\partial_a \det N = 2q_a \det N$ and vanishes initially; no irregularity appears, because the commutant of a regular block forbids a z^{-2} bulk term and the z^{-1} flatness equation then forces the surviving pole to vanish; and the residue of the sheared block moves by conjugation alone, so

its characteristic polynomial is constant. For the cubic that polynomial is $\rho^2+\rho+5/36$ with roots $-1/6, -5/6$, so the primitive-sixth multiplicity is 2 at every bulk parameter, including a specialized one. No receiver, no bulk gauge, and no convergence question enters, because specialization is substitution into constants. Section 10 lists what the earlier versions got wrong; readers of those versions should start there.

1 The assertion at issue

References are to Section 4 of the draft, *Framed formal monodromy.* I name statements by their semantic labels as well as by number, because the edits proposed in Section 9 renumber the section: Definition 4.1 (`def:framed-sixth-multiplicity`), Definition 4.2 (`def:pro-laurent-gauge-group`), Lemma 4.3 (`lem:pv-base-change`), Lemma 4.5 (`lem:formal-base-shift`), Proposition 4.7 (`prop:framed-operations`), Lemma 4.9 (`lem:divisor-tagging`).

Lemma 4.3 ends with the sentence

> *If the gauge uses only integral powers of z at every finite level, this conjugacy preserves the original-z-disc frame,*

whose proof is the single sentence "the deck action fixes every integral power of z at each finite level, so the compatible gauge is single-valued for the framed turn." Lemma 4.5 then *defines*

$$M^{\mathrm{bulk}} = G\, M_{\mathrm{f},V}^{\mathrm{small,shifted}} G^{-1} \tag{4.1c}$$

and Proposition 4.7 concludes that "each center target summand has the framed monodromy of its specialized small connection." The conclusion does not follow from the display: conjugation preserves a characteristic polynomial for trivial reasons, but says nothing about whether the conjugated matrix is the intrinsic turn operator on horizontal solutions of the gauged module. What is missing is a transport lemma together with an algebra in which the turn acts on solutions.

2 The lemma over a differential field

Notation for Exp_V, $K = \mathbf{C} \oplus V$ and Ω_V is the draft's, from (4.0) and (4.0a). Recall that Exp_V is defined on $(K, +)$, not on Ω_V.

The ground field

Let Γ be a totally ordered abelian group and let

$$\mathcal{H} = \Omega_V(\!(\Gamma \oplus \mathbf{Z}e_z)\!)$$

be the Hahn field of series with well-ordered support and coefficients in Ω_V, the value group ordered lexicographically with the e_z-coordinate *last*, as in the draft. Write $z = x^{e_z}$ and let θ multiply the coefficient at (γ, k) by k; this is a derivation because k is additive on the value group, and $\theta z = z$. Its constants are

$$\mathcal{C} := \mathcal{H}^{\theta=0} = \Omega_V(\!(\Gamma)\!).$$

Taking $\Gamma = 0$ gives $\mathcal{H} = \Omega_V((z))$. For $e \geq 1$ put $\mathcal{H}_e = \Omega_V((\Gamma \oplus \frac{1}{e}\mathbf{Z}e_z))$, a degree-$e$ extension containing $w = x^{e_z/e}$, and $\theta_w = e\theta$, so $\theta_w w = w$. All residues below are taken with respect to θ_w on the ramified side; the draft's $M_{\mathrm{RS},V}$ uses the same convention, and the two differ by the factor e if one is careless.

The solution algebra and the turn

Definition 1. Assume the exponential factors under consideration have *constant* coefficients, that is $\varphi \in w^{-1}\Omega_V[w^{-1}]$. The *universal formal solution algebra* $\mathcal{U}_e$ is the $\mathcal{H}_e$-algebra generated by symbols

$$w^\rho \ (\rho \in K), \qquad \ell, \qquad E(\varphi),$$

subject to $w^\rho w^{\rho'} = w^{\rho+\rho'}$, $w^n =$ the element w^n of $\mathcal{H}_e$ for $n \in \mathbf{Z}$, $E(\varphi)E(\varphi') = E(\varphi + \varphi')$, $E(0) = 1$, with

$$\theta_w(w^\rho) = \rho\, w^\rho, \qquad \theta_w(\ell) = 1, \qquad \theta_w\big(E(\varphi)\big) = \theta_w(\varphi)E(\varphi).$$

The two hypotheses buried in that definition — constant coefficients in φ, and exponents ρ in K rather than in Ω_V — are not bookkeeping. The first is false for the modules the draft transports and is discussed at Remark 5; the second is forced because Exp_V is defined only on K, so a module whose residues leave K needs Exp rebuilt over a larger field.

Lemma 2 (Structure)**.** *Fix representatives $\Lambda \subset K$ for $K/\mathbf{Z}$ with $0 \in \Lambda$. Then $\mathcal{U}_e$ is free as an $\mathcal{H}_e$-module with basis $\{w^\rho \ell^m E(\varphi)\}$, $\rho \in \Lambda$, $m \geq 0$. In particular $\mathcal{U}_e \neq 0$ and $\mathcal{H} \subset \mathcal{U}_e$.*

Proof. One cannot argue that $K/\mathbf{Z}$ is torsion-free; it is not, since it contains $\mathbf{Q}/\mathbf{Z}$, and the rational classes are exactly the ones ν_6 counts. Instead use base change. The group algebra $\mathcal{H}_e[K]$ is free over $\mathcal{H}_e[\mathbf{Z}] = \mathcal{H}_e[t, t^{-1}]$ on any set Λ of coset representatives, and $\mathcal{U}_e$ is obtained from it by the base change $\mathcal{H}_e[t, t^{-1}] \to \mathcal{H}_e$, $t \mapsto w$, a surjection with kernel $(t - w)$ generated by a nonzerodivisor. Base change of a free module is free on the image of the same basis. The crossed multiplication $w^\rho w^{\rho'} = w^n w^{\rho''}$ is well defined, and the derivation is consistent with the integer identification because w^n sits at e_z-exponent n/e and θ_w multiplies it by n. The factors $E(\varphi)$ form the group algebra of a torsion-free group and ℓ is a polynomial variable. $\square$

Lemma 3 (The turn)**.** *There is a unique automorphism σ of $\mathcal{U}_e$ with*

$$\sigma(w) = e^{2\pi i/e}w, \quad \sigma(w^\rho) = \mathrm{Exp}_V(\rho/e)\, w^\rho, \quad \sigma(\ell) = \ell + \tfrac{2\pi i}{e}, \quad \sigma\big(E(\varphi)\big) = E(\sigma\varphi),$$

acting on $\mathcal{H}_e$ by multiplying the coefficient at $(\gamma, j/e)$ by $e^{2\pi i j/e}$. It fixes $\mathcal{H}$, hence $\mathcal{C}$, pointwise, and commutes with θ_w.

Proof. The action on $\mathcal{H}_e$ multiplies coefficients by a character of the value group, hence is a support-preserving ring automorphism, and is the identity on $\mathcal{H}$ because there $j \in e\mathbf{Z}$. Multiplicativity in ρ is the homomorphism property of Exp_V on $(K, +)$, and the two prescriptions agree on integral exponents because $\mathrm{Exp}_V(n/e) = e^{2\pi i n/e}$ by (4.0a). Commutation with θ_w is not a diagonality argument, since neither $\sigma(\ell) = \ell + 2\pi i/e$ nor $\theta_w(\ell^m) = m\ell^{m-1}$ is diagonal; it is the direct check $\sigma\theta_w(\ell^m) = m(\ell + \frac{2\pi i}{e})^{m-1} = \theta_w\sigma(\ell^m)$, together with the diagonal cases. $\square$

Lemma 4 (Constants)**.** *Under the constant-coefficient hypothesis of Definition 1, $\mathcal{U}_e^{\theta_w = 0} = \mathcal{C}$.*

Proof. Let v be the Hahn valuation of $\mathcal{H}_e$, additive because the coefficient field is a field; θ_w multiplies coefficients by integers and never enlarges a support, so $v(\theta_w c) \geq v(c)$.

Write $x = \sum c_{\rho,m,\varphi} w^\rho \ell^m E(\varphi)$ in the basis of Lemma 2 with $\theta_w x = 0$; the equation splits over φ. For $\varphi \neq 0$, let M be the top ℓ-degree in that block; for each ρ, $\theta_w(c_{\rho,M}) + \rho c_{\rho,M} + \theta_w(\varphi) c_{\rho,M} = 0$. Every monomial of $\theta_w(\varphi)$ has Γ-part 0 and strictly negative e_z-part, so $v(\theta_w(\varphi)c) < v(c) \leq v(\theta_w c + \rho c)$ for $c \neq 0$ and the lowest term cannot cancel; hence $c_{\rho,M} = 0$, and descending in M kills the block.

For $\varphi = 0$ the coefficient equations are $\theta_w(c_{\rho,m}) + \rho c_{\rho,m} + (m+1)c_{\rho,m+1} = 0$. Descending in m: the top-degree coefficient satisfies the homogeneous equation $\theta_w(c) + \rho c = 0$, and on each monomial this reads $(j+\rho)c = 0$ with $j \in \mathbf{Z}$, so $c = 0$ unless $\rho \in \mathbf{Z}$, that is unless $\rho = 0$. For $\rho = 0$ the top coefficient is θ_w-constant and the next equation gives $Mc_{0,M} = -\theta_w(c_{0,M-1})$, whose right side has zero component in e_z-degree 0; comparing that component gives $c_{0,M} = 0$ for $M \geq 1$. What survives is $\rho = 0$, $m = 0$, $\varphi = 0$ with θ_w-constant coefficient, and the θ_w-constants of $\mathcal{H}_e$ are those supported in e_z-degree zero, namely $\mathcal{C}$. □

Remark 5 (The constant-coefficient hypothesis is necessary, and fails in the application). Allow φ a Novikov coefficient of positive weight, $\varphi = x^\gamma w^{-1}$ with $\gamma \in \Gamma$, $\gamma > 0$ — the shape of every exponential factor the draft actually meets, since the eigenvalues λ of $c_1\star$ have positive Novikov weight. Then $v(\theta_w \varphi) > 0$ in the coefficient-dominant order and the domination argument reverses. Worse, the equation $\theta_w(c) = \theta_w(\varphi)c$ is solved by

$$c = \sum_{n \geq 0} \frac{(-1)^n}{n!} x^{n\gamma} w^{-n},$$

whose support $\{(n\gamma, -n/e)\}_{n \geq 0}$ is increasing, hence well ordered, so $c \in \mathcal{H}_e$ and $cE(\varphi)$ is a θ_w-constant outside $\mathcal{C}$. Lemma 4 is therefore false in that regime, and with it Proposition 6. The structural reason is worth stating on its own: *in the draft's coefficient-dominant order,* $\exp(-\lambda/z)$ *is already a unit of the coefficient field*, so adjoining $E(\varphi)$ as a free symbol duplicates an existing element and inflates the constants. In that order the exponential factors are not symbols at all, and the entire irregularity of the formal solution sits in its divergent unit part, which is exactly where the rate comparison of Section 3 lives.

Framed operators and transport

Proposition 6 (The framed operator is the matrix of the turn)**.** *Let D be a differential module over $\mathcal{H}$, free of rank n, with connection matrix Ω, and let $Y \in \mathrm{GL}_n(\mathcal{U}_e)$ satisfy $\theta Y = -\Omega Y$. Then $\mathrm{Sol}(D) = \{v : \theta v + \Omega v = 0\} = Y\mathcal{C}^n$, free of rank n and σ-stable; the matrix M with $\sigma(Y) = YM$ lies in $\mathrm{GL}_n(\mathcal{C})$ and is well defined up to $\mathrm{GL}_n(\mathcal{C})$-conjugacy. For D over $\Omega_V(\!(z)\!)$ in Levelt–Turrittin form, M is the draft's framed operator: in a solution basis adapted to one deck orbit of length d the matrix of σ is block-cyclic with the draft's intertwiners A_i as blocks, and the cyclic-block determinant gives $\chi(X) = \det(X^d I - U)$ with $U = A_{d-1} \cdots A_0$, which is (4.0b).*

Proof. From $\theta(Y^{-1}) = Y^{-1}\Omega$ one gets $\theta(Y^{-1}v) = Y^{-1}(\theta v + \Omega v)$, so horizontality of v means $Y^{-1}v$ has θ_w-constant entries, and Lemma 4 identifies those with $\mathcal{C}$. Since σ commutes with θ and fixes Ω, it preserves $\mathrm{Sol}(D)$; so $M = Y^{-1}\sigma(Y)$ has entries in $\mathcal{C}$ and is invertible, being a product of invertibles, with inverse $\sigma(Y^{-1})Y$. Two solution matrices differ on the right by an element of $\mathrm{GL}_n(\mathcal{C})$, which conjugates M. □

Theorem 7 (Frame transport)**.** *Let D, D' be as in Proposition 6, with solution matrices Y, Y', and let $G \in \mathrm{GL}_n(\mathcal{H})$ satisfy $\Omega' = G\Omega G^{-1} - \theta(G)G^{-1}$. Then GY is a solution matrix for D' and $\sigma(GY) = (GY)M$. Consequently*

$$M_{\mathrm{f},V}(D') = C_0^{-1} M_{\mathrm{f},V}(D)\, C_0 \qquad \textit{for some } C_0 \in \mathrm{GL}_n(\mathcal{C}),$$

so the two framed operators have the same characteristic polynomial in $\mathcal{C}[X]$ and the same multiplicity at every root of unity.

Proof. $\theta(GY) = \theta(G)Y - G\Omega Y = -\Omega'(GY)$. Since σ fixes $\mathcal{H}$ pointwise, $\sigma(G) = G$, so $\sigma(GY) = G\sigma(Y) = GYM$. Writing $Y' = (GY)C_0$ with $C_0 \in \mathrm{GL}_n(\mathcal{C})$ gives $M' = C_0^{-1}MC_0$. □

Remark 8 (Which object is $M_{\mathrm{f},V}$). The framed operator is an automorphism of the horizontal module; its matrix depends on a choice of *solution* frame and is well defined up to conjugacy over the constants. The conjugate GMG^{-1} appearing in the draft's (4.1c) is not the matrix of the turn in any solution frame — Y' differs from GY by a constant matrix, not by G — but the matrix of the transported automorphism read in the *module* frame of D'. Both readings are legitimate, and they give the same characteristic polynomial, but the draft should say which it means, since conflating them is exactly the step at issue.

Remark 9 (The hypothesis on G is the whole content). Since σ fixes $\mathcal{H}$ pointwise, $\sigma(G) = G$ is automatic once G has entries in $\mathcal{H}$; what matters is that G lies in $\mathcal{H}$ and not in $\mathcal{U}_e$. No hypothesis-free statement exists: Y itself is a gauge over $\mathcal{U}_e$ carrying D to the trivial module, whose framed operator is the identity. The rank-one case shows what survives: if $g \in \mathcal{H}^\times$ and $\theta(g)/g = \delta$, then each monomial gives $(k - \delta)g_{(\gamma,k)} = 0$, so $\delta \in \mathbf{Z}$; a gauge over $\mathcal{H}$ shifts residues only by integers, which Exp_V kills.

The scissors: these hypotheses have no common instance in the application

Theorem 7 is true, and in the application it is empty. Its two hypotheses pull in opposite directions.

- If $\Gamma = 0$, so $\mathcal{H} = \Omega_V(\!(z)\!)$, then Levelt–Turrittin supplies solution matrices, but the pro-Laurent comparison gauge has unbounded negative z-order and is not in $\mathrm{GL}_n(\mathcal{H})$.

- If $\Gamma \neq 0$ with the draft's coefficient-dominant order, the gauge lies in $\mathcal{H}$, but $\mathcal{H}$ is not a formal Laurent series field over its constants and Levelt–Turrittin is unavailable. Concretely, the e_z-component of the Hahn valuation is not a valuation on $\mathcal{H}$: for $f = x^{(0,0)} + x^{(1,-100)}$ and $g = -x^{(0,0)}$ its values on f and g are 0 while its value on $f + g$ is -100. So there is no Newton polygon, no slope filtration and no Turrittin algorithm. Nor can one descend to the bounded-z-order elements: $f = 1 - zx^{-\gamma}$ has inverse $-\sum_{k\geq 1} z^{-k}x^{k\gamma}$, so they do not form a field.

An abstract Picard–Vessiot ring for a module over $\mathcal{H}$ does exist, with constants $\mathcal{C}$ after passing to the divisible hull. So the missing ingredient is *not* a solution algebra. It is a canonical Levelt–Turrittin normal form over $\mathcal{H}$, singling out which element of the differential Galois group deserves to be called the turn. That is the precise shape of the remaining problem, and it also shows why a purely valuation-theoretic argument cannot supply it.

3 The receiver

The draft's receiver, in the proof of Proposition 4.7, is

$$\mathscr{R}_j = \Omega_{V,j} \otimes_{H_{0,j}} H_{\mathrm{tot},j}, \qquad H_{\mathrm{tot},j} = \mathbf{C}(\!(\Gamma_j^{\mathrm{coeff}} \oplus \mathbf{Z}e_z)\!). \tag{4.4c}$$

Why the tensor product cannot simply be dissolved

The map $a \otimes h \mapsto a \cdot h$ into the Hahn field with $\Omega_{V,j}$ coefficients is not well defined: the copy of $H_{0,j}$ inside $\Omega_{V,j}$ consists of degree-zero constants, the copy inside $H_{\mathrm{tot},j}$ of monomial series, and the tensor product exists precisely to identify them. Nor can the two copies be assigned different jobs, with module coefficients read as constants and gauge bookkeeping as monomials: the gauge relation $\Omega' = G\Omega G^{-1} - \theta(G)G^{-1}$ ties G and Ω through the same Novikov variables, so one copy is forced. With the constants reading the gauge leaves the field; with the monomial reading the exponential factors become Γ-supported and Remark 5 applies.

Two things should also be said about $\mathscr{R}_j$ itself. It is a tensor product of two field extensions and need not be a domain, and the draft reads characteristic polynomials and algebraic multiplicities in $\mathscr{R}_j[X]$; that deserves an explicit remark, which the draft partly gives by proving each factor injects.

A better receiver

Extend the Γ-valuation of $H_{0,j}$ to its algebraic closure and then to $\Omega_{V,j}$, and take the field of series $\sum_k c_k z^k$ with $c_k \in \Omega_{V,j}$ whose support $\{(v_\Omega(c_k), k)\}$ is well ordered in the chosen order. This has a single copy of the coefficients, is a field rather than a tensor product of two fields, hosts the pro-Laurent gauge, and hosts the exponential factors as honest units. It is strictly better than (4.4c). It does not by itself supply Levelt–Turrittin, and it does not remove the rate criterion below.

The two rates

Weight a monomial of coefficient weight d and z-power k by $a\,d + b\,k$, $a, b > 0$, and let ε be the minimal weight of a generator of J_j.

- *Pro-Laurent gauges gain one filtration unit per unit of z-pole.* By the recursion (4.1a) and the draft's own conclusion, G_α has no power of z below $-|\alpha|$ and a term of bulk degree m lies in $F^m B$; every monomial of J_j^N has weight at least N. Their weights grow like $m(a-b)$, which together with the draft's finite-below argument gives well ordered support when $a > b$.

- *Splitting gauges lose $w(\Delta\lambda)$ per unit of z-power.* This is the draft's own construction (4.9d): at order n the block off-diagonal coefficient is removed by a Sylvester equation $[K_0, A_n] = B_n$ with K_0 the leading matrix, so one division by an eigenvalue difference per order, and $w(A_n) \geq -n\,w(\Delta\lambda)$. The draft's displayed cubic entry $-14/(81r^2)$ at z-order two exhibits the rate: one factor of r^{-1} per unit of z-power. Three caveats. After a ramification $z = w^e$ the divisions occur once per w-order, so the loss is $e\,w(\Delta\lambda)$ per unit of z-power; at the cubic the draft proves the exponential blocks are unramified in z, so $e = 1$ there. The claim that the regular-singular normalization costs nothing needs the homogeneity argument that residue eigenvalues are rational, verified in the cubic block where $\det L_s = (s+1/6)(s+5/6)$. And $w(\Delta\lambda)$ is not controlled by $w(\lambda)$ when the Picard rank exceeds one, since w and the quantum grading are different functionals and leading terms can cancel.

Both conditions hold together iff

$$\boxed{e\,w(\Delta\lambda) < \varepsilon}$$

The criterion is not an artifact of choosing an order

For $C := e\,w(\Delta\lambda)/\varepsilon \geq 1$ the obstruction is not about well ordering at all. The z^n-coefficient of the single product GP of a pro-Laurent gauge with a splitting gauge is $\sum_m g_m p_{n+m}$, whose terms have weight at least $m(1-C)-nC$; for $C \geq 1$ infinitely many terms have bounded weight, so that coefficient is an infinite sum in no completion, ordered or not. Any repair that forms this product must therefore establish $C < 1$. A repair that never forms it — see route 2 below — is not bound by this.

The criterion depends on a normalization the draft may choose

This is the practical point, and it reverses the recommendation of the first version of this memo. The draft fixes $w(u) = 1$ and $w(s_{j,\ell}) = 1$ independently of L, and sets $w(Q^{i_*d}u^{\rho_C\cdot d}) = L(H\cdot i_*d)+\rho_C\cdot d$. Hence $\varepsilon = 1$ always, while the weights of the Novikov generators grow linearly in L. The criterion is therefore

$$e\big(L(H\cdot i_*d_0) + \rho_C\cdot d_0\big) < c_1\cdot d_0,$$

using that the eigenvalues of $c_1\star$ are homogeneous of quantum degree one when Q^d has degree $c_1\cdot d$ — a convention I have checked against the draft's own low-dimensional vanishing proof, where the Q^β-component of $c_1\star$ maps H^i to $H^{i+2-2c_1\cdot\beta}$, and against its explicit cubic block, where K_0 has eigenvalues $\pm 6r$, 0, 0 with $r = (3q)^{1/2}$ and $c_1\cdot\ell = 2$. Consequences:

- Under the draft's instruction to choose L "so large that" separation holds, the criterion fails at essentially every comparison with two distinct exponential factors — including the cubic endpoint.
- Under the smallest admissible L it can hold. For a projective bundle with $c_1(V)\cdot\ell = 0$ one may take $L = 1$, giving $w(Q^\ell) = 1 = \varepsilon$ and $w(\Delta\lambda) = w(q^{1/2}) = 1/2$.
- Independently of L: separation forces $w \geq \varepsilon$ on every generator of J_j, and the image of Q^{d_0} is a generator, so $c_1\cdot d_0 \geq 2$ is necessary under any admissible weight. Index one fails outright.

So the cheapest available repair is to choose the separating weight minimally rather than large, and to check the resulting inequality case by case. The first version of this memo asserted the opposite, on the false ground that the ratio was scale-invariant; rescaling w does move both rates equally, but L is not a rescaling, because the bulk generators' weights are pinned at one.

4 The iterated receiver

The criterion above is a consequence of one architectural choice, and a reader of the second version of this memo proposed the alternative: the draft puts the intrinsic Novikov variables of the endpoint and the new completion variables into a single ordered group, which is what gives an eigenvalue difference positive weight. Nothing forces that. Treat the intrinsic Novikov fraction field as *coefficients*, and complete only in the new exceptional and bulk directions afterwards.

Two facts make the proposal concrete, and both are visible in the draft.

- *The unbounded negative z-order is bulk-driven, not Novikov-driven.* In the recursion (4.1a) the coefficient G_α has no power of z below $-|\alpha|$, where α indexes the *bulk* parameter η, and the filtration depth is acquired by substituting $\eta \in F^1B$. A completion in the bulk directions alone therefore already hosts the pro-Laurent gauge; the Novikov directions are not needed for it.

- *With the Novikov directions as coefficients, the pathology disappears and the criterion is satisfied rather than evaded.* Eigenvalues of $c_1\star$ become constants of weight zero, so $\exp(-\lambda/z)$ is not a unit of the receiver but a genuine exponential symbol, the constant-coefficient hypothesis of Definition 1 holds, and Remark 5 evaporates. Since $w(\Delta\lambda) = 0$, the inequality $e\,w(\Delta\lambda) < \varepsilon$ holds trivially, and the product GP of Section 3 converges coefficientwise.

The shifts that do involve the exceptional Novikov root do not force those directions back into the filtration. The rank-one unit twist attached to $-(c-1)\lambda_j$, the fixed divisor shift, and the ambient target's $\tau^\circ = O(q^{-1})$ all enter Lemma 4.5 as its a_0 and a_2° terms, which are required only to be *constant in* the positive filtration, not to lie in it. With the Novikov directions as coefficients, $\exp(-a_0/z)$ is an honest exponential symbol rather than the object the draft must argue about separately.

What the sources actually permit

I have checked this against the pinned sources rather than left it as a question, and the answer is positive for the comparison and negative for the gauge.

Iritani's Theorem 5.18 states Ψ as an isomorphism of $\mathbf{C}[z]((q^{-1/s}))[[Q,\widetilde{\tau}]]$-modules, with the change of variables defined over $\mathbf{C}((q^{-1/(r-1)}))[[Q]]$. So the ring is *Laurent* in the exceptional root and *formal* in the ambient Novikov variables and the bulk. Two consequences.

The comparison survives. Inverting the Novikov monomials and completing in the exceptional and bulk directions is a ring homomorphism out of that ring. Ψ, its inverse, and the two inverse identities are module maps over it, so they base-change along it unchanged, and the framed operators' entries lie in the coefficient field, which injects. Nothing in the transport step needs Iritani's proof, only his statement, so the formality of his ring in Q is not by itself an obstacle.

The gauge does not. The source's own reconstruction writes $\tau(\widetilde{\tau}) = \tau^\circ + t(\widetilde{\tau})$ and $\varsigma_j(\widetilde{\tau}) = \varsigma_j^\circ + s_j(\widetilde{\tau})$, with $t(\widetilde{\tau}) \in H^*(X)((q^{-1}))[[Q,\widetilde{\tau}]]$ and s_j of the same shape over $H^*(Z)$. The initial terms are harmless: τ° and ς_j° are by definition the values at $Q = \widetilde{\tau} = 0$, hence carry no Novikov dependence at all, and the exceptional direction remains a completion direction as the draft already has it. But t and s_j vanish only at $Q = \widetilde{\tau} = 0$, so at $\widetilde{\tau} = 0$ they retain pure-Novikov terms. Those have to be removed by the normalized gauge of Lemma 4.5, and that construction converges by topological nilpotence of the bulk shift. Inverting the Novikov directions destroys exactly that.

So the iterated receiver is not free, and the decisive question is now small and graded rather than general:

> Is the pure-Novikov part of the coordinate shift — t and s_j restricted to $\widetilde{\tau} = 0$ — confined to $H^0 \oplus H^2$?

If it is, the string and divisor equations absorb it into the a_0 and a_2° terms of Lemma 4.5, which are required only to be constant in the positive filtration, and the iterated receiver goes through for the ambient summand with $w(\Delta\lambda) = 0$. If it is not, the Novikov directions must stay inside the filtration, eigenvalue differences reacquire positive weight, and the criterion of Section 3 returns.

The grading answers this, and the answer is negative in general. Iritani's Theorem 5.18(5) states that $\tau(\widetilde{\tau})$ and $\varsigma_j(\widetilde{\tau})$ are homogeneous of degree 2, and his conventions are $\deg Q^d = 2c_1(X) \cdot d$ and $\deg \tau_i = 2 - \deg \phi_i$. A pure-Novikov term $Q^d\phi$ therefore satisfies

$$\deg \phi = 2 - 2\, c_1(X) \cdot d,$$

so $c_1 \cdot d = 1$ contributes in H^0, $c_1 \cdot d = 0$ in H^2, and $c_1 \cdot d = -1, -2, \dots$ in $H^4, H^6, \dots$; classes with $c_1 \cdot d \geq 2$ cannot contribute at all. Hence the required confinement to $H^0 \oplus H^2$ follows exactly when every contributing effective class satisfies $c_1 \cdot d \geq 0$, and it is not automatic: Iritani's theorem is stated for an arbitrary smooth projective X, and after several blowups in a weak factorization there is no reason for the intermediate ambient variety to have nef anticanonical class.

So the iterated receiver does not by itself remove arbitrary intermediate fourfolds from the problem. It removes them under a semipositivity hypothesis on the ambient, which the weak factorization does not supply. This is worth stating plainly, because it is the second proposed repair that fails on inspection, and it fails for a reason the draft would have inherited silently.

For the center the same analysis applies, with the additional and unchanged difficulty that its Novikov map is the noninjective one, which is why divisor tagging exists.

The scope consequence is the reason to try this first. If the ambient endpoint is handled this way, no inequality is needed for arbitrary intermediate fourfolds — the case that looked least tractable. What remains is the center summands, and there the draft's own low-dimensional classification is restrictive: for nef canonical class it produces an integral-z gauge to a regular-singular connection, so there is no exponential separation at all and $w(\Delta\lambda) = 0$ again; the remaining minimal cases are $\mathbf{P}^1$, $\mathbf{P}^2$, ruled surfaces, and point blowups. Proving primitive-sixth vanishing directly for those, in the coordinates Iritani's specialization actually produces, is a smaller problem than a general transport theorem.

5 The framing route, checked against the sources and closed

The second version of this memo proposed the following route and called it the probable repair. I state it as it was proposed, then report the three checks, then state what survives. The checks were run against arXiv:2411.02266 as cached, arXiv:2307.03696v4, and arXiv:2307.13555v3; the full verdicts, locators and file hashes are in the companion report *The three framing-compatibility checks, run against Iritani's decomposition.*

The route as proposed

Everything above treats the pro-Laurent bulk gauge of Lemma 4.5 as unavoidable. It may not be. Hinault, Yu, Zhang and Zhang work with F-bundles on the formal u-disc, where a framing is a regular flat connection and, in a framing-flat trivialization, the meromorphic connection takes the normal form

$$\nabla_{u\partial_u} = u\partial_u + u^{-1}K + (\mu D + H).$$

Their Theorem 4.34 characterizes when two connections of that shape are gauge-equivalent under

$$\Phi(u) \in \mathrm{GL}(H[[u]]), \qquad \Phi_{ij}(u) \in k[c_1, \dots, c_r][[u]],$$

with Φ then unique given its value at $u = 0$; the conditions are that the leading operators K be conjugate by a parameter-polynomial ϕ, that the two μ agree, and that the diagonal entries of H agree after conjugation modulo the parameter ideal. Their applications are uniqueness of the decomposition for projective bundles and for blowups, complementing the existence theorems of Iritani–Koto and Iritani.

The shape of that gauge is exactly what this memo has been missing. A gauge in $\mathrm{GL}_n(K[[u]])$ has only nonnegative integral powers of the loop coordinate, adjoins no root of u, and is therefore fixed by the turn $u \mapsto e^{2\pi i}u$. Taking $\Gamma = 0$, so that $\mathcal{H} = \Omega_V(\!(u)\!)$ and Levelt–Turrittin is available, Theorem 7 applies with no further work:

$$\Phi \in \mathrm{GL}_n(K[[u]]), \quad \nabla' = \Phi\nabla\Phi^{-1} - \theta(\Phi)\Phi^{-1} \implies M_{\mathrm{f},V}(\nabla') \text{ and } M_{\mathrm{f},V}(\nabla) \text{ conjugate.}$$

No Hahn receiver, no product GP, and no inequality $e\,w(\Delta\lambda) < \varepsilon$, provided the decomposition of Proposition 4.7 can be identified with an F-bundle isomorphism of the class Theorem 4.34 governs. That identification, the matching of the $u = 0$ datum, and the separation of the center specialization were the three checks.

Check one: the identification does not exist as posed

It fails three times over, and the first two are independent of the draft.

Domain. Section 4.4 of Hinault–Yu–Zhang–Zhang is titled *Equivalence of F-bundles over a point*, and both of its applications evaluate at the closed point $q = t = 0$ of the base, the large-radius limit; the proof of their Theorem 5.16 opens by observing that there the quantum product reduces to the classical cup product, so that the leading operator K_{split} of their (5.18) is a cup-product operator and hence nilpotent. The draft's ν_6 is not evaluated there. It is read where the eigenvalues of $c_1\star$ separate — at the cubic, where the block carries $r = (3q)^{1/2}$ and the indicial roots $\pm 1/6$, $\pm 5/6$. At $q = 0$ that separation collapses and the exponential blocks the framed operator is built from do not exist. Their Theorems 5.20 and 5.24 do reconstruct the isomorphism over the whole base from its restriction at the limit point, but that is uniqueness of a map whose existence is Iritani's and Iritani–Koto's theorem; it is not a transport of framed monodromy between base points.

Hypotheses, for the blowup. Section 4.4 assumes the generalized eigenspaces of K all have the same dimension, and the matrix calculus of Definition 4.33 and Theorem 4.34 rests on the resulting splitting $H_0^{\oplus m} \cong H$. The blowup decomposition is $H^*(X) \oplus \bigoplus_{i=1}^{m-1} H^*(Z)[-2i]$, whose summands differ in dimension whenever $\dim H^*(Z) \neq \dim H^*(X)$; the authors write a general endomorphism there as a matrix with $\Phi_{1,1} \in \mathrm{End}(H^*(X))$ and $\Phi_{i,i} \in \mathrm{End}(H^*(Z))$, so the mismatch is explicit in their own notation. Their blowup statement, Theorem 5.22, does assert existence of the isomorphism, and Theorem 5.24 gives the corresponding uniqueness; the point is not that it is unproved but that it is not *derived from* Theorem 4.34, whose hypotheses do not cover the unequal block sizes, and that its existence is referred to the earlier decomposition work. So Theorem 5.22 supplies no leverage of its own on the blowup half of Proposition 4.7 — equation (4.3), the half the headline theorem cannot do without — beyond what Iritani's theorem already gives. The projective-bundle half is genuinely covered: there the eigenspaces are m copies of $H^*(X)$ by their Lemma 5.8, matching Section 4.4, and Theorem 5.16 is proved.

Purpose. Even where Theorem 4.34 applies, its conclusion compares the source at its base point with the target based at a shifted class $\Delta(a)$, pinned by their (5.17) for projective bundles and (5.23) for blowups. That shift is exactly what the pro-Laurent gauge of Lemma 4.5 exists to undo. The route relocates the bulk displacement rather than removing it.

The two normalizations do agree on where the displacement sits, which is a useful check on the draft's reading of the sources. In their limit-point coordinates the eigenvalue $\lambda_i \in \mathbf{C}[c_1, \ldots, c_m]$ is cohomology-valued, so $\Delta(a)$ acquires components in $H^{\geq 4}$ from the higher Chern classes, while the H^2 components cancel identically — their Lemma 5.8(2) gives $m\lambda_i \equiv m\xi^{i-1} - c_1 V$, cancelling the $c_1 V$ on the right of (5.17) — and the H^2 part of $\Delta(a)$ is left free. In Iritani's coordinates the same data reads $\varsigma_j^\circ = -(c-1)\lambda_j + h_{Z,j} + O(q^{-1/(c-1)})$ with λ_j a *scalar* multiple of $q^{1/(c-1)}$, so the H^0 part is that scalar, the H^2 part is $h_{Z,j}$, and everything of degree at least four has moved into the tail. Same content, different bookkeeping; the draft's split of ς_j° into unit twist, fixed divisor, and positive-filtration tail is right. The tail is irreducible: for a trivial rank-two bundle all $c_i(V)$ vanish and their $\Delta(a)$ is purely $H^0 \oplus H^2$, yet Iritani–Koto's ς_j° still carries its $O(q^{-1/r})$ tail, so even $X \times \mathbf{P}^1$ does not escape the bulk gauge.

Check two: the $u = 0$ datum, which passes

Their Theorems 5.20(2) and 5.24(2) determine the base map from its restriction to the base point up to a multiplicative constant in each logarithmic direction; the proof of their Proposition 4.31 shows why, since df is recovered through $\Psi(d \log p_i) = d \log f_i$ and integrating a logarithmic form leaves one constant. That ambiguity is a character of the Novikov lattice: $q \mapsto cq$ is $Q^d \mapsto c^{E \cdot d} Q^d$, which is the divisor substitution (4.1) of the draft with a_2° a logarithmic multiple of the dual class. The draft already proves such a substitution is a z-constant coefficient automorphism and leaves ν_6 unchanged, and already checks it is well defined on the image monoid. The bundle map carries no ambiguity at all once its restriction is fixed.

The ambiguity also never has to be tolerated, because the sources pin it. Iritani–Koto's (5.11) gives $\varsigma_j^\circ = r\lambda_j + [z^{-1}] \log\big(q^{c_1(V)/(rz)} F_j(1)\big)$ in closed form and their (5.12) gives Φ_j°; Iritani's Section 5.8.1 fixes $\Psi|_{Q=\tilde{\tau}=0}$ from his Lemma 5.13 and the explicit λ_j. One manuscript sentence citing those, and noting that any residual constant is a divisor substitution already covered by (4.1), settles this point for good, whichever repair eventually lands.

Check three: the center specialization, which stays separate

It stays separate more firmly than the check anticipated. Their A-model F-bundle is built throughout on the one-variable Novikov projection $q^\beta \mapsto q^{\beta \cdot \omega}$ of their (2.21), and in the blowup application the center copies are the maximal F-bundles of Z associated to $\sigma^* \omega_X$ — the Novikov direction pulled back from the ambient variety, which is the draft's specialization in a coarser form. No statement of theirs can therefore certify that a specialized center module carries an intrinsic framed monodromy. Divisor tagging remains the route, in the current manuscript, for identifying the specialized center contribution with the intrinsic invariant; route 3 of the list below bypasses that identification instead, by proving specialized vanishing directly in low dimension.

One genuine gain came out of it. Their Lemma 2.24 proves that the collapsed potential determines the full genus-zero potential, by expanding in the H^2 bulk directions and separating curve classes by their intersection numbers with an H^2 basis, under the finiteness of their Assumption 2.22. That is the mechanism of `lem:divisor-tagging`, in print, in a closely related setting. It does not close that lemma — it works at the level of the potential over a base retaining all H^2 directions, whereas the draft needs the separation at the level of a framed operator at a fixed bulk point, and the draft's collapse is by the pair $(i_*, \rho_C\cdot)$ rather than by a single nef class — but it makes the shape of the argument a cited one.

What survives, and the corrected diagnosis

Two things survive, and both are worth more than the route was.

The comparison isomorphisms need no receiver. Iritani's Ψ is an isomorphism of $\mathbf{C}[z]((q^{-1/s}))[[Q,\widehat{\tau}]]$-modules (Theorem 5.18), and Iritani–Koto's Φ is induced by a $\mathbf{C}[z,q][[Q,\widehat{\tau}]]$-module map and is an isomorphism over $\mathbf{C}[z]((q^{-1/r}))[[Q,\widehat{\tau}]]$ (proof of their Theorem 5.1(6)). Both are polynomial in z with inverses in the same ring, hence lie in $\mathrm{GL}_n(F((z)))$ for F the Laurent coefficient field, hence are fixed by the turn. Theorem 7 applies to them with $\Gamma = 0$, where Levelt–Turrittin is available: no Hahn field, no product GP, no inequality. The receiver problem was never a property of the comparison.

The pro-Laurent gauge is not the draft's choice. This corrects the sentence the second version of this memo built its recommendation on. Iritani–Koto transport along the same object: their reconstruction in Section 5.8 uses $M = \bigoplus_j e^{-\varsigma_j^\circ/z} M_B(\varsigma_j^\circ + s_j; Qq^{-c_1(V)/r})$, described in their footnote as a fundamental solution for the summands with respect to $(Q, s_0, \ldots, s_{r-1})$ but not for q and z, satisfying $M = \mathrm{id} + O(z^{-1})$ and lying in $\mathrm{End}[z^{-1}]((q^{-1/r}))[[Q,s]]$. Per bulk degree it is a *polynomial* in z^{-1} — which is the draft's own bound (4.1a), that G_α has no power of z below $-|\alpha|$ — and it is unbounded only after summing over bulk degrees. The sources meet this object and handle it not by an ordered receiver but by Birkhoff factorization, splitting the z-regular factor Φ from the $\mathrm{id} + O(z^{-1})$ factor M' in $(\Phi^\circ)^{-1} \circ M = M' \circ \Phi^{-1}$. Any repair should be measured against that, not against a receiver.

The residual gap

After the two survivals, the blocker is no longer transport across the comparison. It is one statement:

> Let T be smooth projective with quantum connection ∇, and let $\tau^\bullet$ be a bulk parameter in the positive filtration — Iritani's $\tau^\circ = q^{-1}[Z] + O(q^{-2})$ on the ambient summand, the $O(q^{-1/(c-1)})$ tail of ς_j° together with s_j on a center summand. Show that the framed formal monodromy of $\nabla|_{\tau=\tau^\bullet}$ has the same primitive-sixth multiplicity as $\nabla|_{\tau=0}$.

Everything else in Proposition 4.7 is proved or reduced to a regular gauge, and `lem:divisor-tagging` inherits this gap and nothing else. Section 7 proves this statement whenever the block carrying ν_6 is a rank-two Jordan block that is regular singular after one shearing, which covers the cubic ambient summand. What is still open is the same statement for an arbitrary block, which is what a center-by-center treatment would require.

6 The blockwise route, and what the flatness identities already give

This section carries out the second route far enough to see what it costs. It produces the exact evolution of the leading block operator under bulk motion and proves that multiplicity-one blocks carry no framed monodromy at all; for coalesced blocks it does not reduce the problem, but identifies why the full formal normal form, rather than the compressed pair (U_i, μ_i), is what has to be controlled. Conventions are Iritani–Koto's, from their Theorem 5.1: write $U = E\star_\tau$, $C_a = \phi_a\star_\tau$, and

$$\nabla_a = \partial_a + z^{-1}C_a, \qquad \nabla_{z\partial_z} = z\partial_z - z^{-1}U + \mu,$$

with μ the constant grading operator. Horizontal sections satisfy $z^2\partial_z Y = (U - z\mu)Y$.

The two identities

Lemma 10 (Flatness)**.** $[U, C_a] = 0$ *and* $\partial_a U = C_a + [C_a, \mu]$.

Proof. The first is commutativity and associativity of the quantum product. For the second, expand $[\nabla_a, \nabla_{z\partial_z}] = 0$. The z^{-2} term is $z^{-2}[U, C_a]$ and vanishes by the first identity; the z^{-1} term is $z^{-1}\big(-\partial_a U + C_a + [C_a, \mu]\big)$, using $z\partial_z(z^{-1}C_a s) = -z^{-1}C_a s + z^{-1}C_a z\partial_z s$ and $\partial_a\mu = 0$. There is no z^0 term. □

So bulk motion is not an arbitrary negative-loop gauge. The first identity says C_a commutes with every spectral projector P_i of U, hence preserves each generalized eigenspace H_i with $U|_{H_i} = u_i + N_i$; the second says the motion of U itself is expressed through C_a and μ alone.

The block frame

Let $T_a = \sum_j (\partial_a P_j)P_j$ be Kato's transport operator, so that $\partial_a P_i = [T_a, P_i]$ and $P_i T_a P_i = 0$. It is independent of z and has no pole, so it moves the blocks without touching the loop coordinate. For an operator A write $A_i = P_i A P_i$ and $A_{ij} = P_i A P_j$, and let $D_a A_i = \partial_a A_i - [T_a, A_i]$ be the covariant derivative in this frame. A short computation using $\partial_a P_i = [T_a, P_i]$ gives the general rule

$$D_a A_i = P_i(\partial_a A)P_i + P_i[A, T_a]P_i. \tag{6.1}$$

Theorem 11 (Block evolution)**.** *Assume the eigenvalues u_j of U are pairwise distinct as scalars, so that each difference $u_i - u_j$ is invertible. Then, unconditionally,*

$$D_a U_i = C_{a,i} + [C_{a,i}, \mu_i], \qquad D_a\mu_i = \sum_{j\neq i}\big(\mu_{ij}X_{ji} - Y_{ij}\mu_{ji}\big),$$

where $X_{ji} = P_j(\partial_a P_i)P_i$ and $Y_{ij} = P_i(\partial_a P_j)P_j$ are the unique solutions of the Sylvester equations

$$\mathcal{S}_{ji}(X_{ji}) = -(\partial_a U)_{ji}, \qquad \mathcal{S}_{ij}(Y_{ij}) = -(\partial_a U)_{ij}, \qquad \mathcal{S}_{ji}(X) := (u_j - u_i)X + N_j X - X N_i,$$

with $(\partial_a U)_{ji} = C_{a,j}\mu_{ji} - \mu_{ji}C_{a,i}$ *and* $(\partial_a U)_{ij} = C_{a,i}\mu_{ij} - \mu_{ij}C_{a,j}$. *If* both *blocks are semisimple,* $N_i = N_j = 0$, *then* $\mathcal{S}_{ji}$ *is multiplication by* $u_j - u_i$ *and the formula collapses to*

$$D_a\mu_i = [\, C_{a,i}, S_i \,], \qquad S_i := \sum_{j\neq i} \frac{\mu_{ij}\mu_{ji}}{u_i - u_j}. \tag{6.2}$$

Proof. For U: by (6.1) and Lemma 10, $D_aU_i = P_i(C_a + [C_a,\mu])P_i + P_i[U,T_a]P_i$. Since C_a commutes with P_i, the first term is $C_{a,i} + [C_{a,i},\mu_i]$; since U commutes with P_i, the second is $[U_i, P_iT_aP_i] = 0$. No perturbation formula is used, so this identity holds whatever the nilpotent parts.

For μ: $\partial_a\mu = 0$, so (6.1) gives $D_a\mu_i = P_i[\mu, T_a]P_i$, which is the displayed sum. To evaluate X_{ji}, differentiate $[U,P_i] = 0$ to get $[\partial_aU, P_i] + [U, \partial_aP_i] = 0$ and compress by $P_j(\cdot)P_i$ with $j \neq i$: the first bracket contributes $(\partial_aU)_{ji}$ and the second contributes $U_jX_{ji} - X_{ji}U_i$, which is $\mathcal{S}_{ji}(X_{ji})$. The operator $\mathcal{S}_{ji}$ is invertible because $u_j - u_i$ is invertible and $X \mapsto N_jX - XN_i$ is nilpotent. The same argument with i and j exchanged gives Y_{ij}. When $N_i = N_j = 0$, substituting the two displayed expressions for ∂_aU makes the terms containing $C_{a,j}$ cancel, leaving $\sum_{j\neq i}[C_{a,i}, \mu_{ij}\mu_{ji}]/(u_i - u_j)$. □

Remark 12 (The clean form does not cover the blocks that matter). An earlier version of this memo stated (6.2) under the weaker hypothesis $N_j = 0$ for the *neighbouring* blocks only. That was an error: differentiating $UP_i = (u_i + N_i)P_i$, the off-block part of $\partial_a(N_iP_i)$ contributes $X_{ji}N_i$, so the term $-XN_i$ cannot be dropped when the block i itself is nonsemisimple. By Corollary 14 the only blocks that can carry ν_6 are the coalesced ones, and in the draft's cubic calculation that block has $J_0 = \left(\begin{smallmatrix} 0 & 2 \\ 0 & 0 \end{smallmatrix}\right) \neq 0$. So $N_i \neq 0$ is the case of interest, and (6.2) does not apply to it. The cancellation that produced S_i has to be redone with $\mathcal{S}_{ji}^{-1}$ in place of the scalar $1/(u_j - u_i)$; expanding $\mathcal{S}_{ji}^{-1} = \sum_{k\geq 0}(-1)^k(u_j - u_i)^{-k-1}(L_{N_j} - R_{N_i})^k$ shows the clean commutator is only the leading term of that expansion.

Multiplicity-one blocks carry nothing

This is independent of the deformation and settles, in this normalization, the observation the earlier version of this memo only flagged.

Theorem 13 (Simple blocks are invisible)**.** *Let* $(\cdot,\cdot)$ *be the Poincaré pairing. Then each* μ_i *is anti-self-adjoint for the restriction of the pairing to* H_i*, which is nondegenerate; consequently* $\operatorname{tr}\mu_i = 0$ *and the spectrum of* μ_i *is symmetric about* 0. *If* u_i *has multiplicity one, then* $\mu_i = 0$, *the block's formal solution is* $e^{-u_i/z}$ *times a single-valued formal unit, its framed monodromy is the identity, and it contributes no root of unity to* ν_6.

Proof. Quantum multiplication is Frobenius, $(a \star b, c) = (a, b \star c)$, so U is self-adjoint; distinct generalized eigenspaces of a self-adjoint operator are orthogonal, so each P_i is self-adjoint and the pairing restricts nondegenerately to H_i. The pairing vanishes unless the degrees are complementary, so μ is anti-self-adjoint; hence so is $\mu_i = P_i\mu P_i$ on H_i. An anti-self-adjoint operator has $\operatorname{tr}\mu_i = \operatorname{tr}\mu_i^\dagger = -\operatorname{tr}\mu_i$ and characteristic polynomial even up to sign, giving the trace and symmetry statements. On a one-dimensional space anti-self-adjointness forces $\mu_i = 0$. Formal decoupling from the other eigenspaces by a gauge $g = 1 + zg_1 + \cdots$ can add terms $z\,c_1 + z^2c_2 + \cdots$ to the scalar block, but it does not change its z^0 coefficient, since the only z^0 contribution is $-[g_1, U]_{ii} = [U_i, g_{1,ii}] = 0$ in rank one. So the decoupled block is $z^2\partial_z y = (u_i + z^2c_2 + \cdots)y$, whose solution is $e^{-u_i/z}$ times a formal unit with no $\log z$ and no fractional power: single-valued under the turn. □

Corollary 14. *ν_6 is carried entirely by the blocks of $E\star$ whose eigenvalue has multiplicity greater than one.*

This matches the draft's own cubic computation, where K_0 has eigenvalues $\pm 6r, 0, 0$ and the primitive-sixth exponents come from the repeated 0: the $\pm 6r$ blocks are simple, so by Theorem 13 they contribute the identity, and the draft's indicial roots 1/6 and 5/6 are $\pm 1/6$ modulo $\mathbf{Z}$, exactly the symmetry about 0 that the theorem forces on a block residue.

What is left, exactly

Proposition 15 (Isomonodromy criterion, semisimple blocks only)**.** *Suppose $N_i = 0$ and $N_j = 0$ for all j, so that the decoupled block connection is $\nabla_i = z\partial_z + A_i(z)$ with $A_i(z) = -z^{-1}U_i + \mu_i$ and (6.2) applies. Then $D_a\nabla_i$ is an infinitesimal gauge transformation by $\Theta_a(z) = -z^{-1}C_{a,i} + \beta$ if and only if*

$$[\beta, U_i] = 0 \qquad \textit{and} \qquad [\beta, \mu_i] = [\, C_{a,i}, S_i \,].$$

The z^{-2} and z^{-1} obstructions vanish identically for that choice of leading term; only the z^0 equation remains.

Proof. Infinitesimally, $\delta A_i = [\Theta_a, A_i] - z\partial_z\Theta_a$. With $\Theta_a = z^{-1}\alpha + \beta$ this is $-z^{-2}[\alpha, U_i] + z^{-1}\big([\alpha, \mu_i] - [\beta, U_i] + \alpha\big) + [\beta, \mu_i]$. Under the stated hypotheses Theorem 11 gives $D_a\nabla_i = -z^{-1}\big(C_{a,i} + [C_{a,i}, \mu_i]\big) + [C_{a,i}, S_i]$. Taking $\alpha = -C_{a,i}$ kills the z^{-2} term because $C_{a,i}$ commutes with U_i, and reduces the z^{-1} equation to $[\beta, U_i] = 0$; the z^0 equation is the second displayed condition. □

Why this is not yet a reduction of the coalesced case

Proposition 15 is stated where it is true, and that is not where the problem is. Two things fail for a coalesced block, and the second is the deeper one.

First, by Remark 12, S_i is not the right object once $N_i \neq 0$: the full Sylvester inverse has to be carried.

Second, and independently, the compressed pair (U_i, μ_i) does not determine the block's exponents. Formal decoupling of a nonsemisimple block from the other eigenspaces produces higher powers of z, and for a nilpotent leading term the Turrittin reduction reaches down into them. The draft's own cubic calculation is the demonstration. Its reduced rank-two block is

$$z^2\partial_z \begin{pmatrix} \widetilde{S}_3 \\ \widetilde{S}_4 \end{pmatrix} = \big[J_0 + zD_0 + z^2E_0 + O(z^3)\big] \begin{pmatrix} \widetilde{S}_3 \\ \widetilde{S}_4 \end{pmatrix}, \qquad J_0 = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad D_0 = \begin{pmatrix} -19/18 & 0 \\ 0 & 19/18 \end{pmatrix},$$

with E_0 carrying the entry $-8/81$. The ansatz $\widetilde{S}_3 = z^\rho(1 + O(z))$, $\widetilde{S}_4 = cz^{\rho+1}(1 + O(z))$ gives $\rho = 2c - 19/18$ and $c(\rho+1) = \frac{19}{18}c - \frac{8}{81}$, hence $\rho^2 + \rho + \frac{5}{36} = 0$ and $\rho = -\frac{1}{6}, -\frac{5}{6}$. The z^2 coefficient is not decoration: setting its $-8/81$ entry to zero leaves $c(\rho + 1) = \frac{19}{18}c$, whose solutions are $\rho = 1/18$ with $c = 5/9$, and $\rho = -19/18$ with $c = 0$ — the classes $\pm 1/18$ modulo $\mathbf{Z}$, which are eighteenth roots of unity and contribute nothing to ν_6. So a deformation theory that tracks only U_i and $P_i\mu P_i$ cannot see the primitive-sixth exponents at all, and any proposed evolution theorem must reproduce this example. It is the regression test.

The corrected program

1. Flatness preserves the spectral decomposition of U and gives the exact evolution $D_a U_i = C_{a,i} + [C_{a,i}, \mu_i]$, unconditionally.

2. Multiplicity-one blocks contribute no cyclotomic monodromy (Theorem 13), so only coalesced blocks are in play.

3. For a coalesced block the off-block projector equations are governed by the full Sylvester operator $(u_j - u_i) + L_{N_j} - R_{N_i}$, invertible but not scalar.

4. What must be deformed is the entire formal block normal form $J + zD + z^2E + \cdots$, not the compressed pair, and the deformation theory has to reproduce the cubic 2×2 example above.

A caution on the literature before it is leaned on. The extension results for isomonodromic deformations with coalescing eigenvalues assume, among other hypotheses, that the leading matrix is diagonalizable at the coalescence point. The carrier here is nonsemisimple at leading order — $J_0 \neq 0$ — so those theorems do not apply off the shelf, and the coalescence question stated below is not answered by citing them.

The risk that remains is unchanged in kind and now better located: a coalesced block can *split* under the displacement, the splitting being governed by $D_a U_i$, and if the rank-two block carrying $\pm 1/6$ split into simple blocks those would carry residue 0 by Theorem 13 and ν_6 would drop by two. Proving that the reconstruction displacement does not do that, or does it in a count-preserving way, is the residual problem. Section 7 now does exactly that, by shearing the block before deforming it.

7 The coalesced block is rigid

This section closes the residual gap of Section 5 for a block of the cubic's type. The route is the one Section 6 said was needed: deform the *full* formal normal form rather than the compressed pair. The device that makes it finite is to shear first. After the shearing the block is regular singular, its residue already contains the z^2 coefficient that the regression test showed to be load-bearing, and — this is the point — the bulk connection loses its pole, so the residue moves by conjugation alone.

Setting

Let Λ be the coefficient field of the small connection, containing $r = (3q)^{1/2}$ and its inverse, and let $B = \Lambda[[\tau]]$ be the formal even bulk germ in finitely many even coordinates τ_a. Write the quantum connection in the conventions of Section 6,

$$z\partial_z Y = \left(z^{-1}U - \mu\right)Y, \qquad \partial_a Y = -z^{-1}C_a Y,$$

with $U = E\star_\tau$, $C_a = \phi_a \star_\tau$ and μ constant, so that Lemma 10 holds: $[U, C_a] = 0$ and $\partial_a U = C_a + [C_a, \mu]$. Everything below is formal in τ and needs no completion of the loop coordinate, no ordered receiver, and no bulk gauge.

Standing hypotheses, all verified for the cubic at $\tau = 0$:

(H1) $U(0)$ has an eigenvalue u_0 whose generalized eigenspace H_0 has rank two and on which the nilpotent part is nonzero; every other eigenvalue differs from u_0 by a unit of Λ.

(H2) After the shearing of Step 4 the block is regular singular at $\tau = 0$, that is $f(0) = 0$ in the notation there.

For the cubic, (H1) holds with $u_0 = 0$ and the other eigenvalues $\pm 6r$, and (H2) holds because the draft's D_0 is diagonal.

Step 1: decoupling, and the bulk connection stays block diagonal

Lemma 16. *There is a unique gauge $g = I + O(z)$ over B, block-diagonalizing the z-connection with respect to the generalized eigenspaces of U. In the gauged frame the bulk connection is block diagonal as well, and its leading coefficient on H_0 is $C_{a,0} = P_0 C_a P_0$.*

Proof. Existence and uniqueness of g is the splitting lemma at Poincaré rank one, solved order by order by Sylvester equations whose operators are invertible because the eigenvalue differences are units. For the second statement write the gauged pair as $\widetilde{\mathcal{A}} = z^{-1}U + O(1)$, block diagonal, and $\widetilde{\mathcal{C}}_a = z^{-1}\check{C}_a(z)$ with $\check{C}_a = \check{C}^{(0)} + z\check{C}^{(1)} + \cdots$, and expand the flatness identity

$$\partial_a \widetilde{\mathcal{A}} = -z\partial_z \widetilde{\mathcal{C}}_a + [\widetilde{\mathcal{A}}, \widetilde{\mathcal{C}}_a]. \tag{8.1}$$

Its off-block part in bidegree (i,j), $i \neq j$, has zero left side because $\widetilde{\mathcal{A}}$ is block diagonal. At order z^{-2} this reads $(u_i - u_j)(\check{C}^{(0)})_{ij} = 0$, and at each successive order the same invertible factor multiplies the next off-block coefficient. So every off-block coefficient vanishes. Since $g = I + O(z)$, the leading term is unchanged. □

Step 2: the exponential factor, and the commutant

Put $u_0(\tau) = \frac{1}{2}\operatorname{tr}\big(U|_{H_0}\big)$ and $N := U|_{H_0} - u_0 I$, so N is traceless and $N(0) \neq 0$ is nilpotent. Twisting the block by $\exp(u_0/z)$ removes its exponential factor; this changes neither the regular-singular part nor the framed monodromy. The twist replaces the leading bulk coefficient $C_{a,0}$ by $C_{a,0} - (\partial_a u_0)I$.

Lemma 17. *$C_{a,0} = p_a I + q_a N$ with $p_a, q_a \in B$, and $p_a = \partial_a u_0$. After the twist the leading bulk coefficient on the block is $q_a N$.*

Proof. $[U, C_a] = 0$ gives $[U|_{H_0}, C_{a,0}] = 0$. A 2×2 matrix over a local ring whose reduction is non-scalar is regular, and $U|_{H_0}$ reduces to $u_0(0)I + N(0)$ with $N(0) \neq 0$; the commutant of a regular 2×2 matrix is free of rank two on I and the matrix itself, hence equals $B \cdot I \oplus B \cdot N$. Taking traces in $\partial_a U = C_a + [C_a, \mu]$ on the block gives $2\partial_a u_0 = \operatorname{tr} C_{a,0} = 2p_a$. □

Step 3: the eigenvalue never splits

Theorem 18 (No splitting)**.** $\det N \equiv 0$ *on B. Consequently N is nilpotent at every point of the formal germ, the block keeps a single eigenvalue of multiplicity two, and by (H1) it is a single Jordan block.*

Proof. Write $d := \det N \in B$; by (H1), $d(0) = 0$. From Theorem 11 the block leading operator evolves by $D_a U_0 = C_{a,0} + [C_{a,0}, \mu_0]$, so by Lemma 17

$$D_a N = D_a U_0 - (\partial_a u_0) I = q_a N + q_a [N, \mu_0].$$

For a traceless 2×2 matrix $\operatorname{adj}(N) = -N$ and $N^2 = -dI$, so

$$\partial_a d = \operatorname{tr}\bigl(\operatorname{adj}(N)\, D_a N\bigr) = -q_a \operatorname{tr}(N^2) - q_a \operatorname{tr}\bigl(N[N, \mu_0]\bigr) = 2 q_a d,$$

using $\operatorname{tr}(N[N, \mu_0]) = \operatorname{tr}(N^2 \mu_0) - \operatorname{tr}(N \mu_0 N) = 0$. Now suppose $d \neq 0$ and let $m \geq 1$ be the order of its lowest nonvanishing homogeneous part d_m in τ. The right side of $\partial_a d = 2 q_a d$ has order at least m, while the left side has order $m - 1$ with homogeneous part $\partial_a d_m$. Hence $\partial_a d_m = 0$ for every a, so $d_m = 0$, a contradiction. Note d is basis independent, so no choice of block frame enters. $\square$

This is the falsifier of Section 6 answered: the reconstruction displacement cannot split the block that carries ν_6. It also shows the Kato frame is not needed for this step, since $\det N$ is frame independent.

Step 4: shear

By Theorem 18, $N^2 = 0$ and $N \neq 0$, so $\ker N = \operatorname{im} N$ is a free rank-one submodule; choose a frame (e_1, e_2) of H_0 with $N e_1 = 0$, $N e_2 = \nu e_1$, $\nu \in B^\times$, depending formally on τ, and let Γ_a be the resulting frame connection, which has no pole in z. Apply $S = \operatorname{diag}(1, z)$ and write the sheared pair as

$$z\partial_z \widehat{Y} = \widehat{\mathcal{A}} \widehat{Y}, \quad \widehat{\mathcal{A}} = z^{-1} A_{-1} + A_0 + z A_1 + \cdots, \qquad \partial_a \widehat{Y} = -\widehat{\mathcal{C}}_a \widehat{Y}, \quad \widehat{\mathcal{C}}_a = z^{-1} K_a + G_a + \cdots.$$

Conjugation by S sends $E_{12} \mapsto z E_{12}$, $E_{21} \mapsto z^{-1} E_{21}$ and fixes diagonal matrices, and $-S^{-1} z \partial_z S = -\operatorname{diag}(0, 1)$. Writing the decoupled block connection before shearing as $z^{-1} N + A_0' + z A_1' + \cdots$, one gets

$$A_{-1} = f\, E_{21}, \quad f := (A_0')_{21}, \qquad A_0 = R := \nu E_{12} + \operatorname{diag}\bigl((A_0')_{11}, (A_0')_{22} - 1\bigr) + (A_1')_{21} E_{21}. \tag{8.2}$$

On the bulk side, the leading coefficient $q_a N = q_a \nu E_{12}$ shears *up* to order z^0, and the only surviving pole comes from the $(2, 1)$ entry of the regular part:

$$K_a = k_a E_{21}, \qquad k_a := \bigl(\check{C}^{(1)}_{a,0} + \Gamma_a\bigr)_{21}. \tag{8.3}$$

There is no z^{-2} term, and this is where Lemma 17 does its work: a $(2, 1)$ entry in $C_{a,0}$ would have produced one, and the commutant forbids it.

For the cubic, (8.2) reproduces the draft's own numbers. With $\nu = 2$, $A_0' = D_0 = \operatorname{diag}(-19/18, 19/18)$ and $(A_1')_{21} = -8/81$,

$$f = 0, \qquad R = \begin{pmatrix} -19/18 & 2 \\ -8/81 & 1/18 \end{pmatrix}, \qquad \operatorname{tr} R = -1, \quad \det R = \tfrac{5}{36},$$

so the characteristic polynomial of R is exactly $\rho^2 + \rho + 5/36$ and its roots are $-1/6$ and $-5/6$. The z^2 coefficient enters through $(A_1')_{21}$, so the regression test of Section 6 is met by construction rather than by accident: the object being deformed below is R, which sees it.

Step 5: no irregularity appears, and the bulk connection is regular

Theorem 19 (The sheared block stays regular singular)**.** *$f \equiv 0$ and $k_a \equiv 0$ on B. Hence at every point of the formal germ the block is regular singular with residue R, and in the sheared frame the bulk connection has no pole in z.*

Proof. Expand (8.1) for the sheared pair. The coefficient of z^{-2} is $[A_{-1}, K_a] = [fE_{21}, k_a E_{21}] = 0$, which is vacuous. The coefficient of z^{-1} is

$$\partial_a(f)E_{21} = k_a E_{21} + f\,[E_{21}, G_a] + k_a\,\mathrm{ad}_R(E_{21}).$$

Writing $R = \begin{pmatrix} \alpha & \nu \\ \gamma & \delta \end{pmatrix}$ one computes $\mathrm{ad}_R(E_{21}) = \begin{pmatrix} \nu & 0 \\ \delta-\alpha & -\nu \end{pmatrix}$, whose diagonal is $\mathrm{diag}(\nu, -\nu)$ and does not involve γ. The left side is off-diagonal, so comparing diagonal entries gives $k_a\nu = -f\,\big([E_{21}, G_a]\big)_{11}$, and since ν is a unit, $k_a = f\,h_a$ for some $h_a \in B$. Comparing $(2,1)$ entries then gives

$$\partial_a f = f\Big(h_a\big(1+\delta-\alpha\big) + \big([E_{21}, G_a]\big)_{21}\Big),$$

a linear equation $\partial_a f = f\,w_a$. By (H2), $f(0) = 0$, and the lowest-order argument of Theorem 18 applies verbatim, so $f \equiv 0$ and therefore $k_a \equiv 0$. ☐

Step 6: the exponents are constant

Theorem 20 (Rigidity of the formal type)**.** *$\partial_a R = [R, G_a]$ for every bulk direction a. Hence the characteristic polynomial of R has coefficients in Λ, constant in τ; the formal monodromy of the block is conjugate to $\exp(2\pi i R)$ at every point of the formal germ; and the primitive-sixth multiplicity of the block is constant.*

Proof. By Theorem 19 the sheared pair has $A_{-1} = 0$ and $K_a = 0$. The coefficient of z^0 in (8.1) is $\partial_a A_0 = [A_{-1}, H_a] + [A_0, G_a] + [A_1, K_a]$, which collapses to $\partial_a R = [R, G_a]$. A matrix moving by an infinitesimal conjugation has constant characteristic polynomial: $\partial_a \det(XI - R) = 0$ entrywise. Since $\widehat{\mathcal{A}} = R + O(z)$, the block is regular singular and its formal monodromy is conjugate to $\exp(2\pi i R)$; shearing changed the exponents by integers only, which ν_6 does not see. ☐

Corollary 21 (The residual gap, closed at the cubic)**.** *For a smooth cubic threefold, the framed formal monodromy of the zero block has $\nu_6 = 2$ at every point of the formal even bulk germ, in particular at any bulk parameter $\tau^\bullet$ in the positive filtration. The statement of Section 5 therefore holds for the cubic ambient summand.*

Proof. Combine Theorem 20 with the computation after (8.2): the characteristic polynomial is $\rho^2 + \rho + 5/36$ everywhere, with roots $-1/6, -5/6$, whose monodromy eigenvalues $e^{\mp\pi i/3}$ are primitive sixth roots of unity. Since the coefficients of the characteristic polynomial are constants of Λ, evaluation at a topologically nilpotent $\tau^\bullet$ is legitimate and returns the same polynomial; no convergence of a bulk gauge is involved. ☐

Three remarks

Remark 22 (Why this is not the compressed-pair argument). The deformed object is R of (8.2), not the pair (U_0, μ_0). The shearing has folded the z^2 coefficient of the unsheared block into the residue, so the failure mode of Section 6 — deleting $-8/81$ and moving the exponents to $\pm 1/18$

— cannot occur here: that deletion changes R itself, hence its characteristic polynomial, and Theorem 20 freezes exactly that polynomial.

Remark 23 (Relation to Cai's Proposition 6). Cai states the $\pm 1/6$ exponents for the *big* quantum connection and proves it by constructing $M = I + \sum M_n t^n$ with $\partial_{t_i} M = -z^{-1} P_i M$, so that $M^{-1} \nabla_{t_i} M = \partial_{t_i}$ and $M^{-1} \nabla_z M$ is bulk independent. That gauge is precisely the pro-Laurent object of Lemma 4.5 and of Iritani–Koto's Section 5.8: per bulk degree it is polynomial in z^{-1}, unbounded only after summing bulk degrees. Over $\Lambda((z))[[t]]$, with the bulk kept formal, that argument is sound and proves the same conclusion, because M is fixed by the turn and Theorem 7 applies with the bulk variables as formal parameters. What it does not do by itself is license evaluation at a *specialized* bulk parameter with Novikov coefficients, which is where the receiver discussion of Section 3 came from. The argument above avoids the gauge altogether and produces a τ-constant characteristic polynomial, so specialization is a substitution into constants.

A shorter route to the endpoint, in the sources' own terms

Two paragraphs of Katzarkov–Kontsevich–Pantev–Yu's Example 6.21 bear directly on this and should be on the record, because between them they suggest a route to the one-stabilization theorem that needs neither this memo's transport lemma nor its rigidity theorem.

First, they compute the atomic composition of a smooth cubic threefold at the hyperplane point: two one-dimensional atoms for the nonzero eigenvalues of $\mathrm{Eu}\,\star$, and one atom $\alpha(X)$ for the eigenvalue 0. That is the block structure used throughout this memo, in their language. They add that "the Witt algebra argument from Remark 3.14 shows that there is no further splitting in a neighborhood of b" — which is Theorem 18, asserted rather than proved, and their Remark 3.14 introduces its finer clauses with "more fine results can be proved". Section 7 supplies proofs of both the splitting statement and the finer one, for this block type, from flatness alone.

Second, and more consequentially, they separate $\alpha(X)$ from a curve atom not by formal monodromy but by a Serre automorphism. They observe that $\alpha(X)$ looks like the atom of a smooth projective genus-five curve, then note that as a Hodge atom enhanced with its Serre automorphism S one has the graded minimal polynomial $S^5 = [3]$, so S cannot have eigenvalue -1 and cannot be the Serre automorphism of a genus-five curve. They conclude that every smooth cubic threefold is irrational, and remark that the same approach applies to other Fano threefolds.

The consequence for the endpoint is worth stating plainly. Their argument is dimension three, where Proposition 5.30 requires excluding only points and curves. The one-stabilization statement is dimension four, where the same proposition requires excluding points, curves *and surfaces*. Everything else — the projective-bundle formula putting two copies of $\alpha(X)$ in $X \times \mathbf{P}^1$, the non-rationality criterion, and the decoration itself — is already theirs. So the endpoint theorem may reduce to one finite classification question: whether any atom of a smooth projective surface can carry a Serre automorphism with $S^5 = [3]$. For a surface with nef canonical class their Lemma 5.24 gives a single atom and Claim 6.15 gives a regular singularity with nilpotent residue after the half-parity gauge; the remaining surfaces are $\mathbf{P}^2$, ruled surfaces and point blowups, which the projective-bundle and blowup formulas reduce to points and curves.

This is not the same invariant as ν_6, and it is not obviously weaker or stronger; it is the sources' own, which is an argument for using it. Two cautions. It buys the Hodge-atom, motive and Serre-enhancement machinery that the earlier blueprint deliberately avoided, so the manuscript would import more than the two decomposition theorems. And their sentence is a worked example, not a theorem with hypotheses; before anything rests on it, the graded minimal polynomial and the surface exclusion both need to be checked at the level of detail this memo has applied to everything else.

Remark 24 (Scope). Steps 2 to 5 used rank two only through the identities $\mathrm{adj}(N) = -N$ and $N^2 = -\det(N)I$ and through the single shearing. The mechanism — commutant of a regular block kills the would-be z^{-2} bulk term, one shearing makes the block regular singular, flatness then forces the bulk pole to vanish — should extend to a single Jordan block of any size with the shearing $\mathrm{diag}(1, z, \ldots, z^{m-1})$, and this is worth doing because it is what a general center statement would need. It does not cover a derogatory block, where the commutant is larger, nor a semisimple block, where the z^{-1} equation instead reads $(I + \mathrm{ad}_R)(C_{a,0} - p_a I) = 0$ and a resonance $\rho_i - \rho_j = -1$ must be excluded before the same conclusion follows.

8 Routes, in the order I would now try them

The framing route has been removed from this list by the checks of Section 5. The blockwise route has now been carried to its conclusion for a block of the cubic's type in Section 7, so what follows is the residue of the list: what the remaining routes are still for.

1. **Blockwise, avoiding transport altogether.** Developed in Section 6 and completed in Section 7 for a single Jordan block of rank two. What remains of this route is generalization, not repair: a single Jordan block of arbitrary size, by the same shearing with $\mathrm{diag}(1, z, \ldots, z^{m-1})$, and the semisimple case with its resonance condition. That generalization is what a statement covering arbitrary centers would need; it is not needed for the cubic endpoint. Signs follow the displayed connection of Iritani–Koto's Theorem 5.1 and should be rechecked against the draft's normalization before anything is written into the manuscript.

2. **Birkhoff.** The identity $(\Phi^\circ)^{-1} \circ M = M' \circ \Phi^{-1}$ of Iritani–Koto's Section 5.8, and its counterpart in Iritani's Section 5.8.2, already factors the bulk transport into a z-regular factor, which Theorem 7 handles, and a factor $M' = \mathrm{id} + O(z^{-1})$. What must be shown is that the second factor does not change the primitive-sixth multiplicity. This is a statement about a $z = \infty$-side gauge acting on a $z = 0$ invariant, so it is not automatic, and it is close to the original difficulty in another form: any proof will have to use where M' comes from, which is the flat quantum connection — that is, the identities route 1 already exploits directly. Worth keeping as a cross-check rather than as the primary attack.

3. **The low-dimensional centers directly**: primitive-sixth vanishing after Iritani's specialization, using the classification the draft already proves, rather than through a general transport statement.

4. **Divisor tagging on its own terms**: whether the divisor equation gives a more explicit transport than the general positive-bulk gauge of Lemma 4.5. Their Lemma 2.24 is the same mechanism in print and is worth citing; it does not close the lemma, for the reasons under check three.

5. **Normalize, then verify.** Replace "choose L so large" by the smallest weight satisfying separation, adopt the single-copy receiver of Section 3, and verify $e\,w(\Delta\lambda) < \varepsilon$ case by case. This is fallback machinery now.

The iterated receiver of Section 4 is not on the list: it needs a semipositivity hypothesis on the ambient variety that weak factorization does not supply. It remains worth having where it applies. Restricting Proposition 4.7 to centers satisfying the criterion is likewise available, at a cost in generality that is the author's call.

The observation that multiplicity-one blocks contribute nothing, flagged in the second version of this memo on the strength of Dubrovin's zero-diagonal fact in canonical coordinates, is no longer a flag: Theorem 13 proves it in this normalization, from self-adjointness of $E\star$ and anti-self-adjointness of μ for the Poincaré pairing. So every route above needs to control only the coalesced blocks.

A separate scope reduction stands on its own, unchanged: a direct quantum Künneth computation for $X \times \mathbf{P}^1$ together with a direct computation for $\mathbf{P}^4$ would leave the blowup step as the only operation formula the headline theorem needs, with the genus-eight corollary waiting. Note that this cannot be had by triviality of the bundle alone — see the end of check one.

9 Placement in the manuscript

Stated by label because these insertions renumber Section 4. Items 1 to 3 and item 5 are unconditional and can be written now; the checks of Section 5 also make item 6 partly unconditional, since the comparison isomorphisms themselves are z-polynomial and the transport lemma applies to them with no receiver. With Section 7 the step that removes the positive-filtration bulk parameter is unconditional as well for the cubic ambient summand, so what is left conditional is the corresponding step at an arbitrary center. A ninth item is therefore owed: the rigidity theorem itself, as a lemma before `prop:framed-operations`, stated for the block type it covers and with its two hypotheses displayed rather than assumed. Nothing here should be written into the manuscript before the sign conventions are rechecked against the draft's normalization.

1. The ground field and solution algebra go immediately before "Apply Levelt–Turrittin after a ramification $z = w^e$", which already refers to a universal formal solution algebra that is never constructed.

2. Lemma 4 and Proposition 6 go after `def:framed-sixth-multiplicity`, converting that definition into a characterization, with Remark 5 stating the hypothesis that makes it available.

3. Theorem 7 becomes `lem:frame-transport`, after the paragraph beginning "The frame matters" and before `def:pro-laurent-gauge-group`.

4. In `lem:pv-base-change`, the asserted final clause and the corresponding sentence of its proof become a reference to `lem:frame-transport` — but only once the receiver repair makes that lemma applicable there, since that lemma lives over the pro-Laurent receiver.

5. At (4.1c) in `lem:formal-base-shift`, one sentence identifying M^{bulk} as the transported framed automorphism, in the module frame, with the solution-frame statement cited.

6. `prop:framed-operations` cites the lemma at its three load-bearing steps and says which field plays the role of $\mathcal{H}$ at each. Two of those steps are now unconditional: the transport across Ψ and across Φ needs only that both are z-polynomial with inverses in the same ring, which their statements give, so the field there is $F(\!(z)\!)$ over the Laurent coefficient field

and no criterion is verified. The step that removes the positive-filtration bulk parameter is the one that waits.

7. A sentence in `prop:framed-operations` fixing the normalization of the coordinate change, citing Iritani–Koto (5.11)–(5.12) and Iritani's Section 5.8.1, and noting that any residual constant in a logarithmic Novikov direction is a divisor substitution already covered by (4.1). This is owed regardless of which repair lands, and is check two of Section 5.

8. `lem:divisor-tagging` carries the same obstruction — it embeds the gauge of `lem:formal-base-shift`, whose negative z-order is unbounded, together with every transported operator, into one ordered Hahn field — and needs whatever repair the receiver gets.

Mechanically, a new theorem-like environment must carry exactly one semantic label and a matching row in the claim map consumed by the manuscript's source-only gate, with an empty declaration list exactly when its coverage is recorded as absent, and the coverage snapshot string in both verification READMEs must be regenerated.

10 Corrections to the earlier versions of this memo

From the fourth version to this one

Nothing in the fourth version is retracted. Section 7 is added, and three of its consequences change earlier readings.

1. Section 6 said the coalesced case was not reduced and named the block-splitting risk as the residual problem. Both are now settled for a rank-two Jordan block: Theorem 18 excludes the splitting and Theorem 20 freezes the exponents. The section is left standing because its diagnosis — that the compressed pair is the wrong object — is what the shearing acts on.

2. The residual gap of Section 5 is closed for the cubic ambient summand by Corollary 21. It remains open as stated for arbitrary smooth projective T, which is a statement about arbitrary blocks, not about transport.

3. Cai's Proposition 6 already asserts the $\pm 1/6$ exponents for the big quantum connection, by a bulk gauge of exactly the pro-Laurent shape this memo scrutinized. That argument is sound with the bulk kept formal; the receiver discussion was always about specialization, not about the formal statement. Section 7 makes the point moot by producing constants.

From the third version to the fourth

Both are in Section 6, and both are at the coalesced blocks, which are the ones that matter.

1. The off-block projector equation was derived as $(\partial_a U)_{ji} + (u_j - u_i + N_j)X_{ji} = 0$, dropping the term $-X_{ji}N_i$. The correct equation is $\mathcal{S}_{ji}(X_{ji}) = -(\partial_a U)_{ji}$ with $\mathcal{S}_{ji}(X) = (u_j - u_i)X + N_jX - XN_i$. The closed form $D_a\mu_i = [C_{a,i}, S_i]$ therefore holds only when the block itself is semisimple, and the blocks carrying ν_6 are exactly the ones where it is not; in the draft's cubic block $J_0 \neq 0$.

2. The block was modelled as $z\partial_z - z^{-1}U_i + \mu_i$. After formal decoupling a nonsemisimple block carries higher powers of z, and its exponents depend on them: in the cubic block the z^2 coefficient's entry $-8/81$ enters $\rho^2 + \rho + 5/36 = 0$, and deleting it moves the exponents from $\pm 1/6$ to $\pm 1/18$. So the isomonodromy criterion stated there is a statement about semisimple blocks and is not a reduction of the residual problem. It is restated with that hypothesis.

The simple-block theorem survives, with one wording repair: the formal solution of a simple block is $e^{-u_i/z}$ times a single-valued formal unit, not times a constant, since decoupling can generate higher z-terms there too. The monodromy conclusion is unaffected.

From the second version to the third

Both changes come from running the three checks against the sources.

1. The framing route of Section 5 was called the probable repair. It is not a repair at all: its governing theorem lives at the large-radius limit point; its blowup case, though stated with existence in Theorem 5.22 and uniqueness in Theorem 5.24, is not derived from Theorem 4.34, whose hypotheses do not cover the unequal block sizes, and its existence is referred to the earlier decomposition work; and the gauge it supplies lands at the shifted base point the bulk gauge exists to undo.

2. The second version wrote that the unbounded negative loop order is the draft's own choice rather than something the comparison theorems impose. That is wrong, and it was the ground for the recommendation. Iritani–Koto transport along the same object, their fundamental solution M of Section 5.8, with the same per-bulk-degree z^{-1}-polynomiality the draft derives in (4.1a).

Two findings were gained in exchange, and they narrow the problem rather than restate it: the comparison isomorphisms need no receiver, and the residual gap is one displacement statement. Both are in Section 5.

From the first version to the second

For readers of the first version, these are the substantive changes.

1. The Constants Lemma was stated without the constant-coefficient hypothesis and is false without it; see Remark 5. The framed-operator characterization inherits the hypothesis.

2. The scissors were only implicit. The transport theorem has no verified instance in the application, and that now appears where the theorem does.

3. The claim that the criterion is scale-invariant, and the inference that the separating parameter L cannot be tuned, were both wrong. Tuning L down is now the first proposed route.

4. The cubic-endpoint bullet compared two different normalizations. Corrected above.

5. The finite-tensor objection to the receiver was wrong twice over: the pro-Laurent gauge lies in the Hahn factor alone, and the splitting gauge's coefficients lie in one finite extension of $H_{0,j}$. It is replaced by the gauge-relation argument.

6. The remark that a tensor product of two fields is not a domain was asserted as a general principle, which is false. The correct statement is that $\mathscr{R}_j$ need not be one.

7. The obstruction is now given in its order-free form, which answers two of the questions the first version left open: no completion helps, and the criterion is not an artifact of the ordered receiver.

8. Smaller repairs: the freeness proof (the relevant quotient group has torsion), the commutation of σ with the derivation, the inverse of M, the range of the exponents, the ramification factor in the splitting rate, and the divisor-tagging observation, which is new.

11 Questions

The first question of the second version — whether the two connections compared in Proposition 4.7 lie in the class Theorem 4.34 governs — has been answered no, three times over, in Section 5. Questions 1 and 2 of the fourth version are answered in Section 7: the object to deform is the residue of the sheared block, which sees the z^2 coefficient by construction, and its characteristic polynomial is constant; and the displacement cannot split the block, because $\det N$ satisfies a linear equation with zero initial value. Question 3 is no longer on the critical path, since the coalescence literature is not used. These remain.

1. Does the argument of Section 7 extend to a single Jordan block of arbitrary size, and to a semisimple block once the resonance $\rho_i - \rho_j = -1$ is excluded? This is what a statement about arbitrary centers, rather than the cubic endpoint, would need.

2. Is hypothesis (H2) — that the sheared block is regular singular at the base point — structural rather than computed? Self-adjointness of $E\star$ and anti-self-adjointness of μ force the $(2,1)$ entry of μ_0 to vanish in the shearing frame, which is suggestive; what is missing is the same statement for the decoupling correction.

3. Does the factor $M' = \mathrm{id} + O(z^{-1})$ of the sources' Birkhoff factorization preserve the primitive-sixth multiplicity? Now a cross-check rather than a route.

4. For the low-dimensional centers, can primitive-sixth vanishing be proved pointwise in the coordinates Iritani's specialization produces, given that the nef-canonical case already yields an integral-z gauge to a regular-singular connection and that Theorem 13 disposes of every multiplicity-one block? A pointwise statement there needs no transport at all, which is the shape the endpoint theorem wants.

5. Which presentation should the manuscript take: the operation formulas with ν_6 transported summand by summand, which needs the generalization in item 1, or the atom formulation, in which the intermediate fourfolds never appear because the weak-factorization bookkeeping happens at the level of atoms as objects?